

CASEMENT, Ireland

WMO: 039670

Lat: 53.300N

Lon: 6.433W

Elev: 319

StdP: 14.53

Time Zone: 1.00 (IRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	26.5	29.3	23.3	17.8	30.3	25.8	20.0	32.1	36.9	47.0	34.1	46.9	5.3	230	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	73.2	63.6	70.2	61.9	67.9	60.7	64.9	70.4	63.4	68.3	62.0	66.7	10.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.8	86.5	67.7	61.3	81.8	65.9	59.9	78.0	64.6	30.1	70.6	28.9	68.4	28.0	66.8	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
31.1	27.3	24.3	DB	21.0	77.8	4.1	3.2	18.0	80.1	15.6	82.0	13.3	83.8	10.3	86.1
			WB	20.4	68.0	4.1	1.9	17.5	69.4	15.1	70.5	12.8	71.6	9.8	73.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	41.6	41.7	43.7	47.0	51.8	56.6	59.9	59.4	55.9	51.0	45.4	42.2	(6)
	DBStd	8.27	5.36	5.12	4.95	4.94	4.74	4.27	3.92	3.78	4.16	4.93	5.48	6.13	(7)
	HDD50	1302	261	235	202	115	34	2	0	0	4	47	155	247	(8)
	HDD65	5588	724	652	660	541	410	254	164	178	276	433	589	707	(9)
	CDD50	1206	2	3	7	24	89	200	308	292	179	80	17	5	(10)
	CDD65	15	0	0	0	0	0	2	7	5	1	0	0	0	(11)
	CDH74	62	0	0	0	0	1	11	34	15	1	0	0	0	(12)
	CDH80	3	0	0	0	0	0	0	2	1	0	0	0	0	(13)
(14) Wind	WSAvg	11.4	13.9	13.3	12.4	10.7	10.4	9.8	9.4	9.9	10.3	11.7	12.1	13.3	(14)
(15) Precipitation	PrecAvg	29.1	2.6	2.0	1.9	2.0	2.3	2.2	2.0	2.5	2.5	3.0	3.0	2.9	(15)
	PrecMax	40.2	4.8	5.0	3.9	4.9	5.9	7.0	6.0	6.8	6.2	7.0	7.2	8.1	(16)
	PrecMin	22.6	0.4	0.3	0.3	0.2	0.6	0.2	0.4	0.1	0.3	0.6	0.5	0.9	(17)
	PrecStd	4.1	1.2	1.2	0.8	1.0	1.1	1.4	1.1	1.3	1.4	1.6	1.6	1.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	56.4	59.5	65.6	70.4	75.4	77.7	75.9	71.7	64.7	59.6	56.7	(19)
		MCWB	52.8	52.1	52.3	55.4	60.3	61.9	65.1	65.4	63.5	60.0	56.6	54.4	(20)
	2%	DB	53.1	53.6	56.0	60.6	66.6	70.7	73.8	71.4	68.0	62.0	57.3	54.7	(21)
		MCWB	51.0	50.5	50.9	53.1	57.9	61.0	64.1	63.6	61.8	58.1	54.7	52.5	(22)
	5%	DB	51.6	51.7	53.7	57.5	63.2	67.8	70.3	68.6	65.4	60.3	55.4	53.0	(23)
		MCWB	49.7	49.4	50.1	51.5	56.0	59.6	62.3	61.8	59.8	56.8	53.0	50.8	(24)
	10%	DB	50.1	49.9	51.8	55.1	60.1	65.1	67.6	66.5	63.2	58.8	53.6	51.2	(25)
		MCWB	48.0	47.5	48.5	50.0	54.1	58.2	60.8	60.6	58.3	55.6	51.4	49.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	53.1	53.9	56.9	62.2	64.7	67.2	67.1	65.4	61.0	57.5	54.7	(27)
		MCDB	54.5	55.0	56.6	63.2	69.7	71.5	74.5	73.3	69.5	63.5	59.0	56.1	(28)
	2%	WB	51.5	51.2	52.0	54.3	59.0	62.3	65.1	64.7	62.8	59.0	55.2	52.8	(29)
		MCDB	52.9	53.2	54.9	59.1	64.7	68.3	71.4	69.3	66.6	61.3	56.8	54.4	(30)
	5%	WB	49.9	49.5	50.5	52.3	56.9	60.6	63.4	63.1	61.1	57.4	53.4	51.1	(31)
		MCDB	51.4	51.4	53.4	56.7	61.9	66.5	68.9	67.4	64.6	59.9	55.1	52.7	(32)
	10%	WB	48.2	47.8	48.7	50.5	55.0	58.9	61.8	61.7	59.1	55.9	51.7	49.4	(33)
		MCDB	49.9	49.9	51.5	54.5	59.3	64.1	66.6	65.8	62.3	58.4	53.5	51.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	10.2	12.0	13.4	13.9	14.0	13.6	13.3	12.8	11.1	9.8	9.4	(35)
		MCDBR	10.2	10.5	13.5	15.9	17.8	17.9	17.3	15.3	15.1	11.3	9.6	10.2	(36)
	5% WB	MCWBR	9.9	9.4	9.9	10.4	11.0	10.8	9.7	9.3	10.0	9.0	9.0	9.7	(37)
		MCWBR	10.3	10.4	12.3	14.6	15.9	16.1	15.4	14.1	13.4	10.7	9.4	10.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.335	0.345	0.372	0.387	0.393	0.382	0.385	0.366	0.359	0.347	0.327	0.323	(40)
		taud	2.258	2.327	2.277	2.268	2.293	2.355	2.351	2.417	2.433	2.415	2.329	2.231	(41)
	Ebn at Noon		195	237	255	267	271	275	272	272	259	235	199	173	(42)
		Edh at Noon	20	26	33	38	39	37	36	32	28	23	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		223	444	791	1213	1539	1603	1456	1226	913	535	281	171	(44)
	RadStd		21	37	82	122	112	162	132	107	51	44	30	21	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page